Problem Statement: Google Maps Business Data Extractor API

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Background and problem

Sales, marketing and research teams need structured lists of local businesses, such as every dental clinic around Dharampeth, Nagpur. Today this data is collected by hand from Google Maps: searching, scrolling, opening each listing and copying name, phone, address and rating into a spreadsheet.

This manual process has clear problems:

- **Slow:** collecting 100 listings takes 2–4 hours of repetitive work.
- **Error-prone:** typos, missed listings and duplicate rows are common.
- **Not repeatable:** every new category or area means starting over from scratch.
- **Not integrable:** other tools (CRM, lead dashboards) cannot request the data automatically.

Objective and proposed solution

Build a backend service with a REST API that takes a **business category** (e.g. "dental clinic") and a **location** (e.g. "Dharampeth, Nagpur"), collects every matching business listed on Google Maps, and returns the results as a downloadable **Excel (.xlsx) file**.

How a request flows through the system:

1. Client calls the API with a keyword and location.
2. Service validates input and geocodes the location to latitude/longitude.
3. Service queries Google Maps data for matching places, following all result pages.
4. For each place, service fetches details (phone, website, hours, rating).
5. Results are cleaned and de-duplicated by place ID.
6. Data is written to an .xlsx file and returned as a download (or a download link for large jobs).

Scope

In scope

- Any business category as free-text input (dental clinic, gym, restaurant, CA firm, etc.).
- Any location as free text (area, city, landmark) with an optional search radius.
- Pagination to collect all available results, not just the first page.
- Excel export with one row per business and a fixed column layout.
- Basic job tracking for long-running searches (status: queued, running, done, failed).

Out of scope (v1)

- Scraping customer reviews text or photos.
- Finding emails or social profiles from business websites.
- A web front-end (API only; a simple UI can follow in v2).
- Scheduled or recurring crawls.

Functional requirements

Input

Field	Type	Required	Example
query	string	Yes	dental clinic
location	string	Yes	Dharampeth, Nagpur
radius_km	number	No (default 5)	3
max_results	number	No (default: all)	200
min_rating	number	No	4.0

Output: Excel columns (one row per business)

Column	Description
Sr. No.	Row number
Business Name	Name as listed on Google Maps
Category	Primary type, e.g. Dentist

Column	Description
Full Address	Formatted address
Area / Locality	Neighbourhood, e.g. Dharampeth
City / Pincode	City and postal code
Phone Number	Local or international format
Website	Business website URL
Rating	Average star rating (1–5)
Total Reviews	Number of reviews
Opening Hours	Weekly schedule
Business Status	Operational / temporarily closed / permanently closed
Latitude, Longitude	Coordinates
Google Maps Link	Direct URL to the listing
Place ID	Unique Google identifier, used for de-duplication

Rules

- Remove duplicate listings by Place ID.
- Leave a cell blank when a field is not available; never fill placeholders.
- Name the file `<query>_<location>_<date>.xlsx`, e.g. `dental_clinic_dharampeth_nagpur_2026-10-14.xlsx`.
- Add a second sheet, "Search Info", recording query, location, radius, timestamp and total count.

API specification

The API uses an async job pattern, because a full search with pagination and detail lookups can take 30 seconds or more.

Method	Endpoint	Purpose
POST	/api/v1/search	Start a search job; returns a job_id

Method	Endpoint	Purpose
GET	/api/v1/search/{job_id}	Check job status and result count
GET	/api/v1/search/{job_id}/download	Download the .xlsx file

Request

```
POST /api/v1/search
{
  "query": "dental clinic",
  "location": "Dharampeth, Nagpur",
  "radius_km": 3,
  "max_results": 200
}
```

Response

```
{
  "job_id": "a1b2c3",
  "status": "queued",
  "status_url": "/api/v1/search/a1b2c3"
}
```

Status response (when done)

```
{
  "job_id": "a1b2c3",
  "status": "done",
  "total_results": 87,
  "download_url": "/api/v1/search/a1b2c3/download"
}
```

Error responses: 400 for missing or invalid input, 401 for a missing API key, 404 for an unknown job_id or unresolvable location, 429 when the rate limit is exceeded, 500 for upstream failures.

Non-functional requirements

- **Performance:** return up to 60 results in under 30 seconds; larger jobs run in the background.

- **Reliability:** retry failed upstream calls up to 3 times with exponential backoff; a partial failure still produces a file with what was collected.
- **Security:** protect endpoints with an API key; keep the Google API key in environment variables, never in code.
- **Rate limiting:** cap each client at a set number of jobs per hour to control cost.
- **Logging:** log every job with query, location, result count, duration and errors.
- **Storage:** delete generated Excel files after 24 hours.
- **Caching:** reuse results for the same query + location within 24 hours to save API cost.

Deliverables

- Working REST API with the three endpoints above.
- Excel output matching the column layout in Functional requirements.
- README with setup steps, environment variables and sample requests.
- Postman collection or Swagger (OpenAPI) docs.

Success criteria

- [ ] Searching "dental clinic" in "Dharampeth, Nagpur" returns a valid .xlsx with all required columns.
- [ ] Results match what Google Maps shows for the same search, with at least 95% coverage.
- [ ] Zero duplicate rows in the output.
- [ ] Invalid input returns a clear error message, not a crash.
- [ ] A 60-result search completes in under 30 seconds.